

Unit Practice Test

Geometric Constructions & Rigid Transformations | 53 points | about 60 minutes

Name: _____

Date: _____

Score: / 53

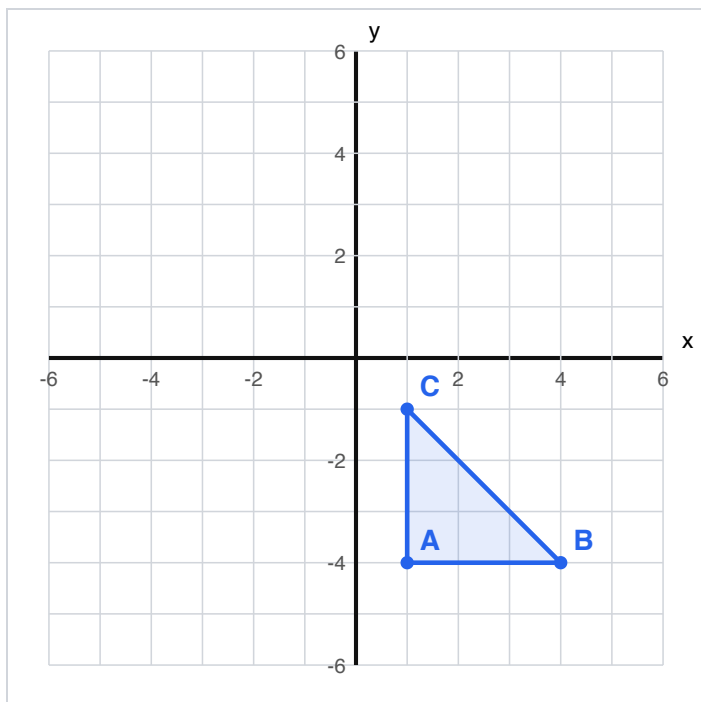
Use pencil, a ruler, and a compass. On grid problems, draw the image, label it with primes (A' , B' , C'), and write the new coordinates beside the grid. Counterclockwise is the positive direction for rotations.

Part 1: Multiple Choice (1 point each)

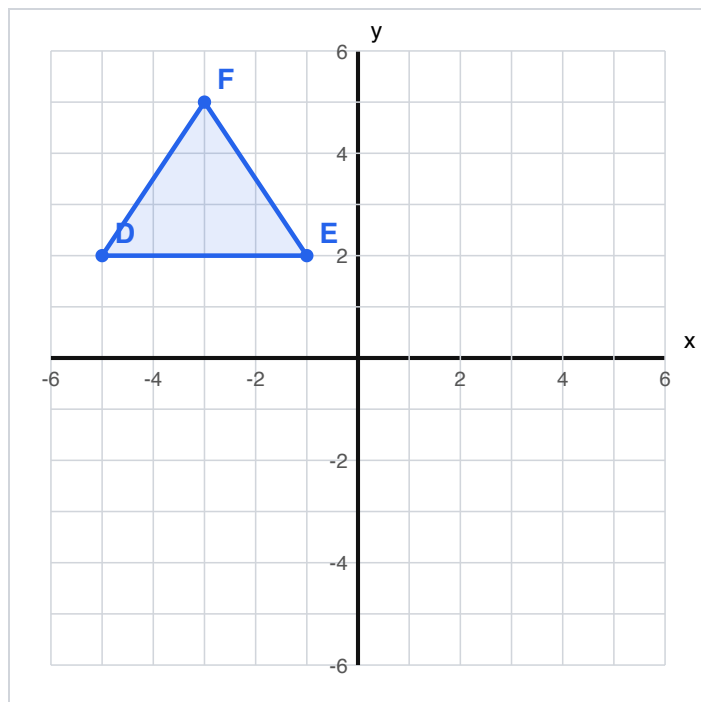
- Which of these is **not** a rigid transformation?
 - Translation
 - Reflection
 - Dilation by a factor of 3
 - Rotation
- A reflection maps $P(-3, 4)$ to $P'(-3, -4)$. What is the line of reflection?
 - the x-axis
 - the y-axis
 - $y = x$
 - $x = -3$
- Rotating $(2, 5)$ by 90° counterclockwise about the origin gives:
 - $(5, -2)$
 - $(-5, 2)$
 - $(-2, -5)$
 - $(5, 2)$
- A translation maps $(-2, 3)$ to $(4, -1)$. Where does it send $(0, 0)$?
 - $(2, 2)$
 - $(-6, 4)$
 - $(6, -4)$
 - $(4, -1)$
- When you construct a perpendicular bisector, you draw arcs of the same radius from both endpoints. Why does that make the crossing points land on the bisector?
 - Equal radii make each crossing point equally far from both endpoints
 - It makes the segment twice as long
 - It makes the arcs parallel
 - It guarantees a 60° angle
- Triangle $A'B'C'$ is the image of triangle ABC after a reflection. Which statement must be true?
 - $AB = A'B'$
 - $\angle A = 2(\angle A')$
 - The area of $A'B'C'$ is double that of ABC
 - The vertices go around in the same direction

Part 2: Draw the Image (2 points each)

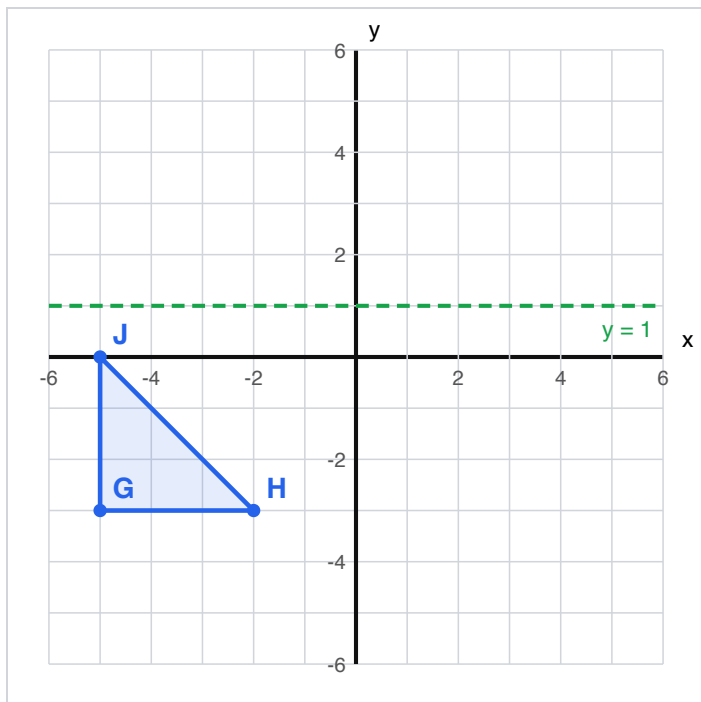
7. Translate triangle ABC by $\langle -3, 4 \rangle$.



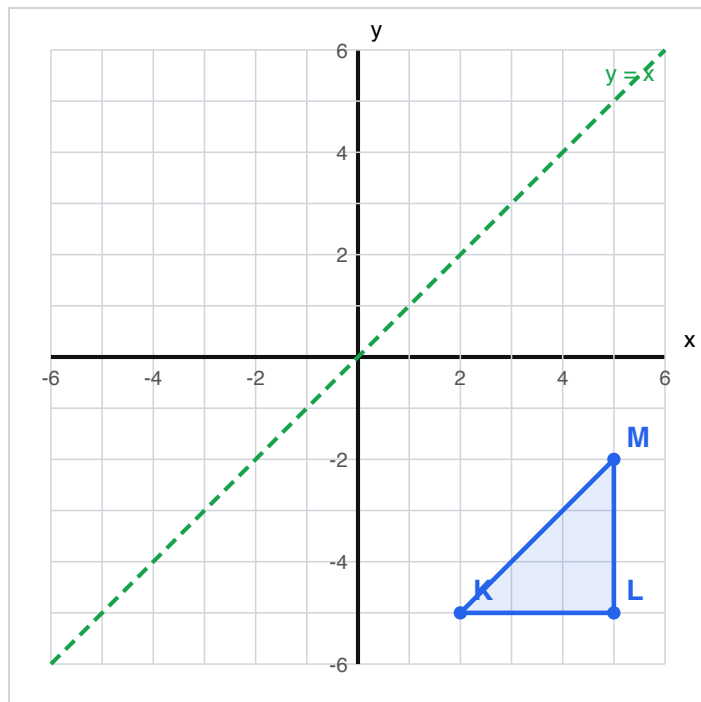
8. Reflect triangle DEF over the **x-axis**.



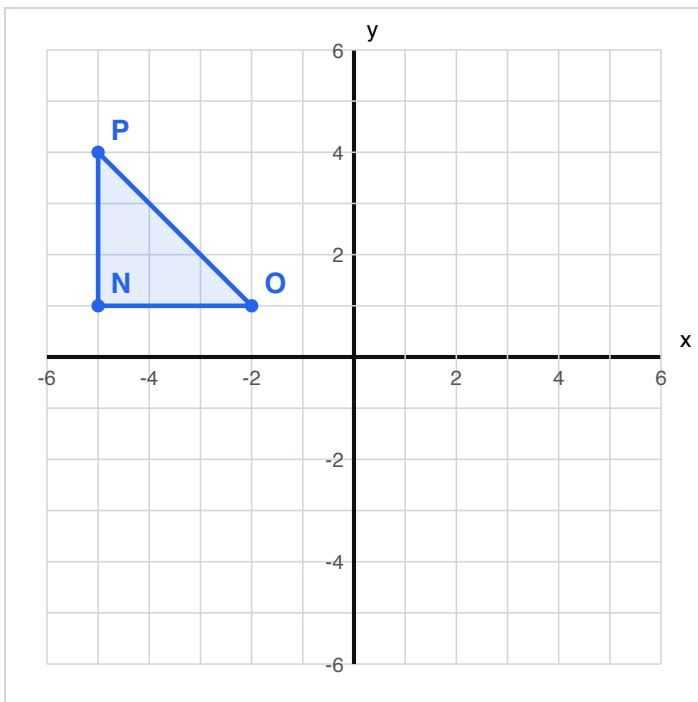
9. Reflect triangle GHJ over the line **$y = 1$** . (Each point and its image are the same distance from the line.)



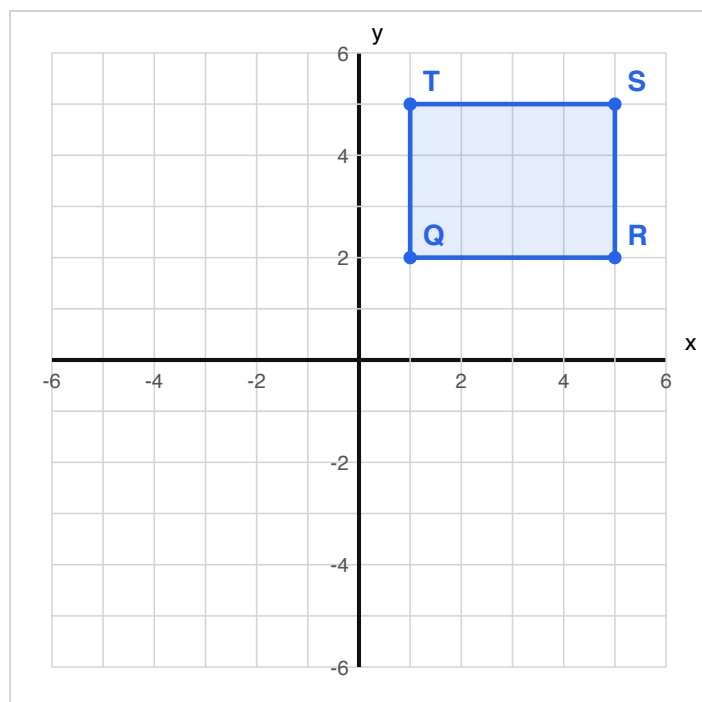
10. Reflect triangle KLM over the line **$y = x$** .



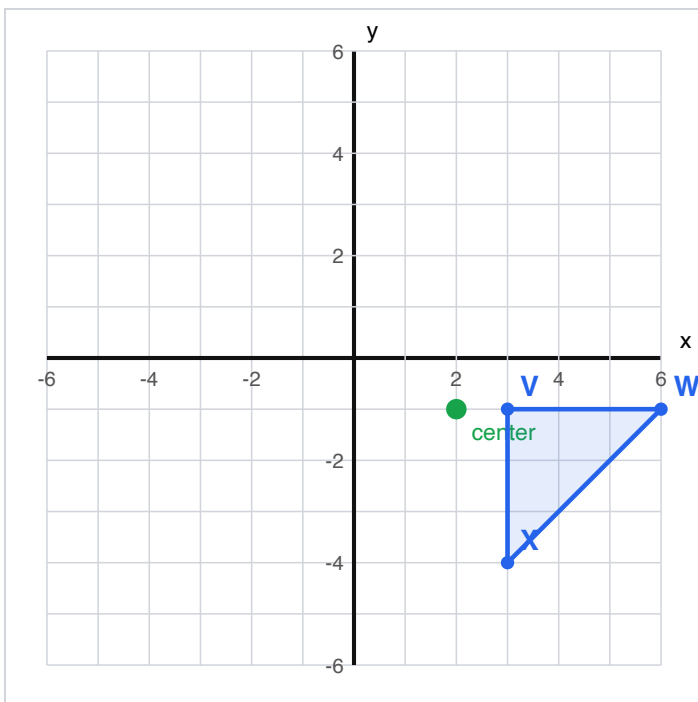
11. Rotate triangle NOP **90° clockwise** about the origin.



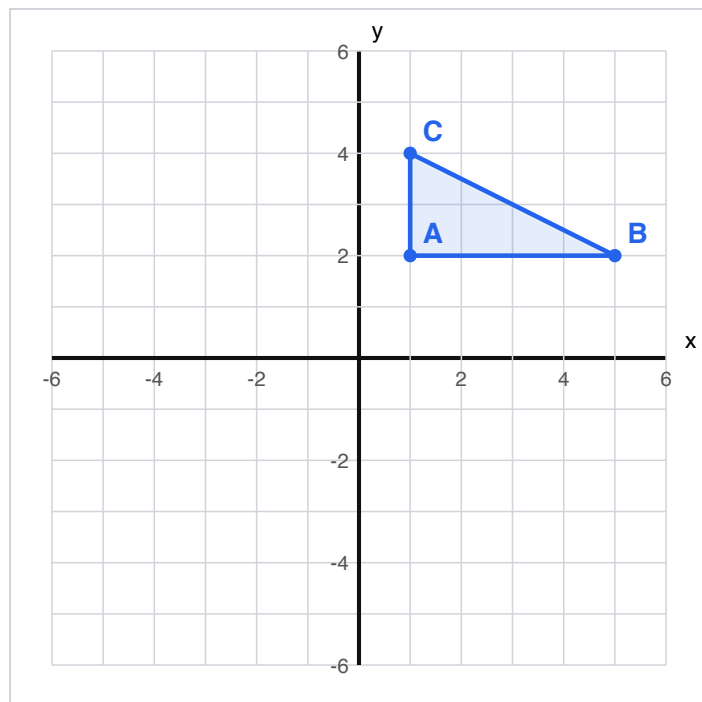
12. Rotate square QRST **180°** about the origin.



13. Rotate triangle VWX **90° counterclockwise** about the point (2, -1), **not** the origin.

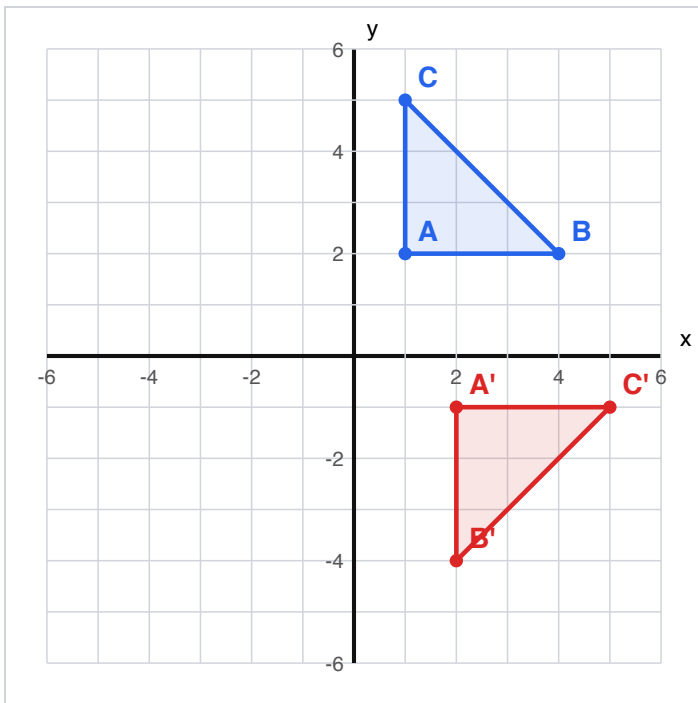


14. Triangle ABC: **first** reflect over the y-axis, **then** translate by $\langle 1, -3 \rangle$.

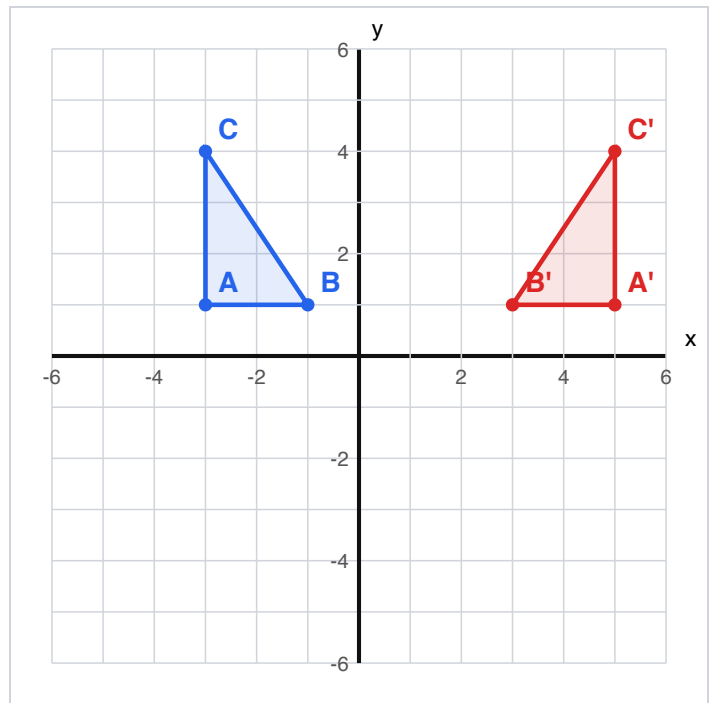


Part 3: Describe the Transformation (2 points each)

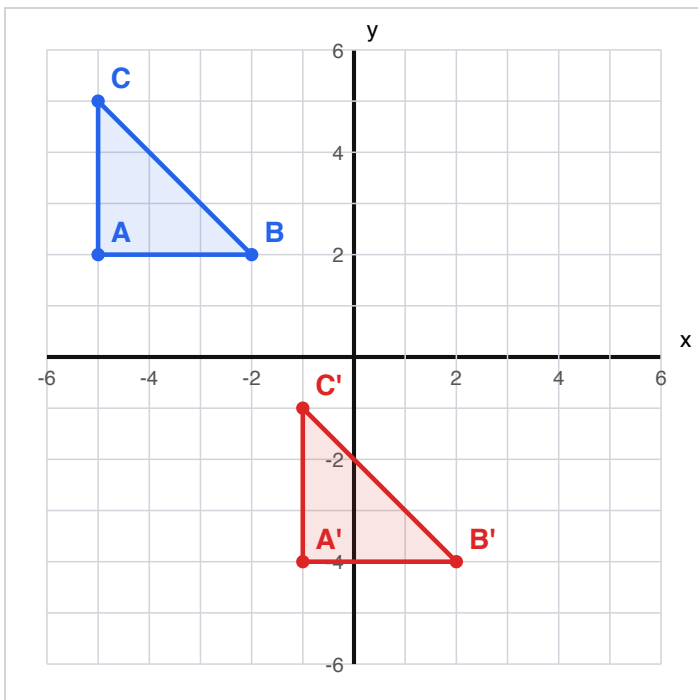
15. Describe the transformation that maps triangle ABC to A'B'C'. Be specific (direction, angle, center).



16. Describe the transformation that maps triangle ABC to A'B'C'. Give the equation of the line.



17. Describe the transformation that maps triangle ABC to A'B'C'. Give the vector.



Part 4: Congruence and Reasoning (2 points each)

18. Triangle ABC has A(0, 0), B(3, 0), C(0, 4). Triangle DEF has D(5, 1), E(8, 1), F(5, 5). Describe a translation that maps ABC onto DEF, and use it to explain why the triangles are congruent.

19. Triangle PQR has P(1, 1), Q(4, 1), R(1, 3). Triangle P'Q'R' has P'(2, 2), Q'(8, 2), R'(2, 6). Could a rigid transformation map PQR onto P'Q'R'? Use side lengths to explain.

20. The point (4, -3) is rotated 90° counterclockwise about the origin and then reflected over the x-axis. Find the final point. Would doing the two steps in the opposite order give the same point? Show work.

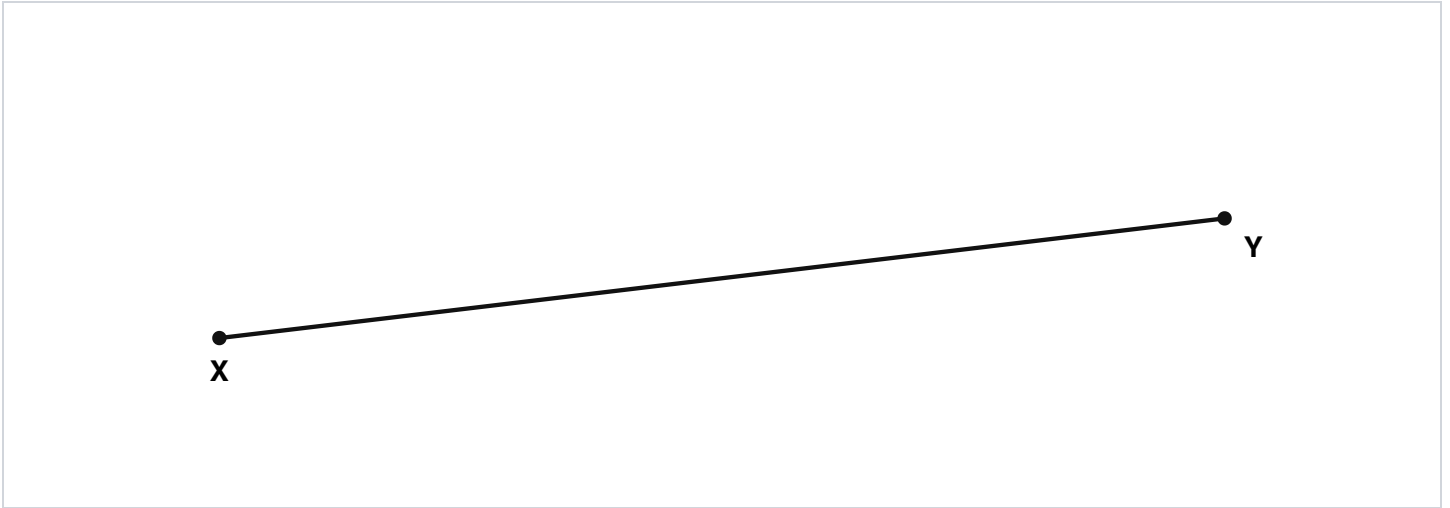
21. Segment AB has endpoints A(-2, 1) and B(2, 4). It is reflected over the y-axis. Find the length of A'B' **without** computing the new coordinates, and explain how you know.

22. A(2, 6) maps to A'(2, 0) and B(-3, 5) maps to B'(-3, 1) under a reflection. Find the line of reflection and check it with both points.

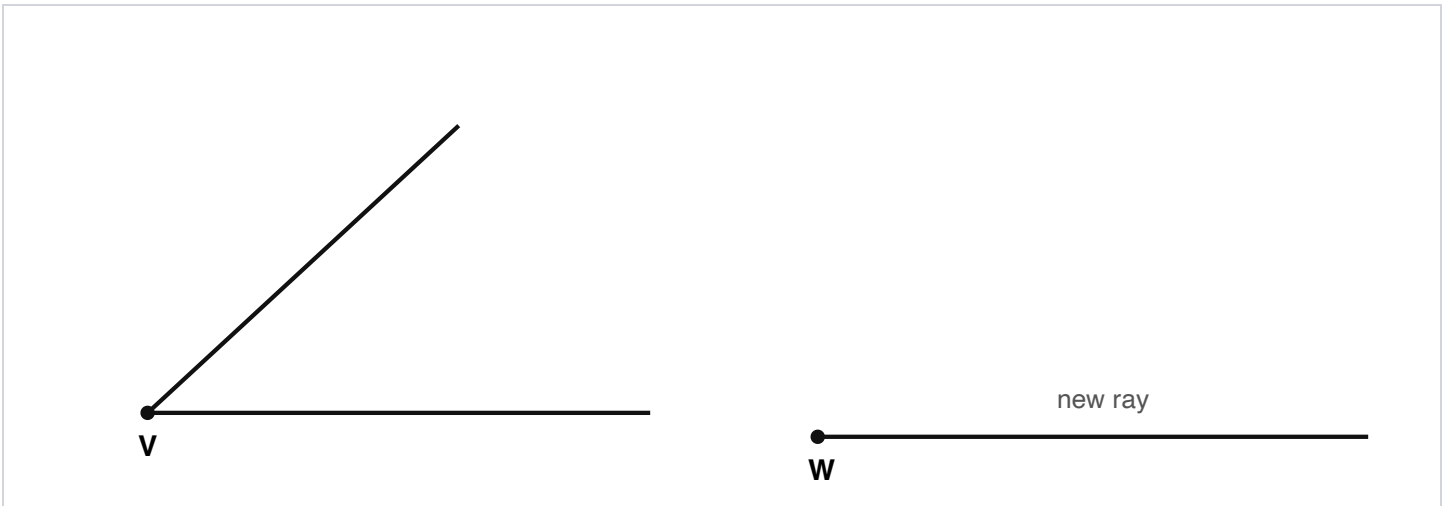
Part 5: Constructions (3 points each: 2 for the construction, 1 for the explanation)

Leave all your construction arcs showing. Write one sentence below each figure explaining why your construction works.

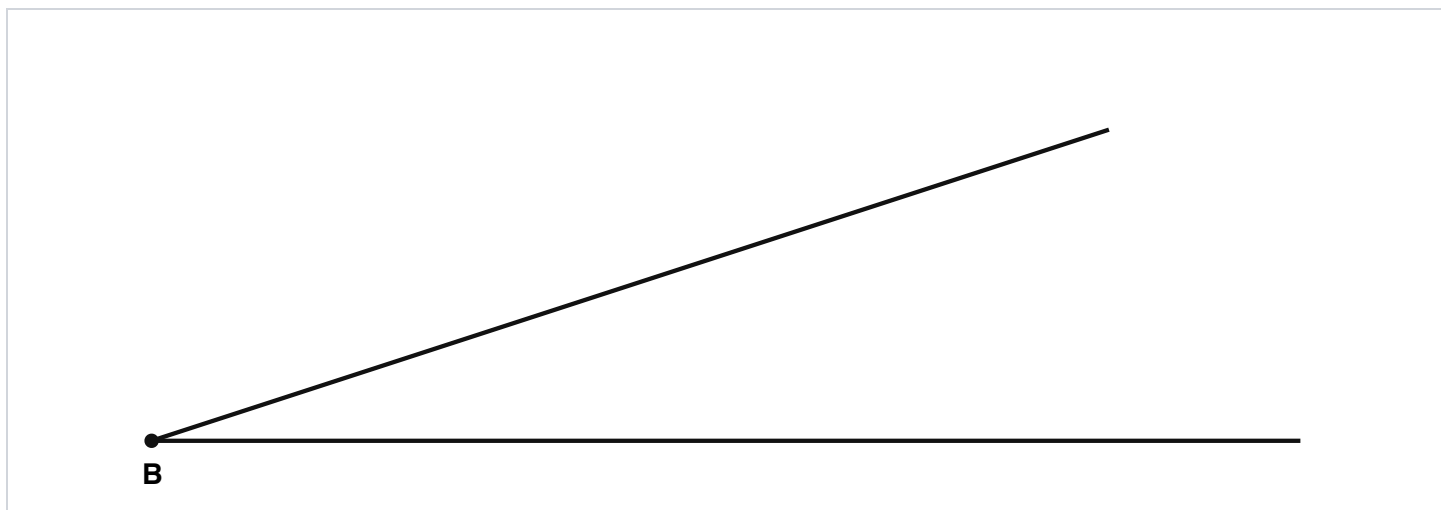
23. Construct the **perpendicular bisector** of XY .



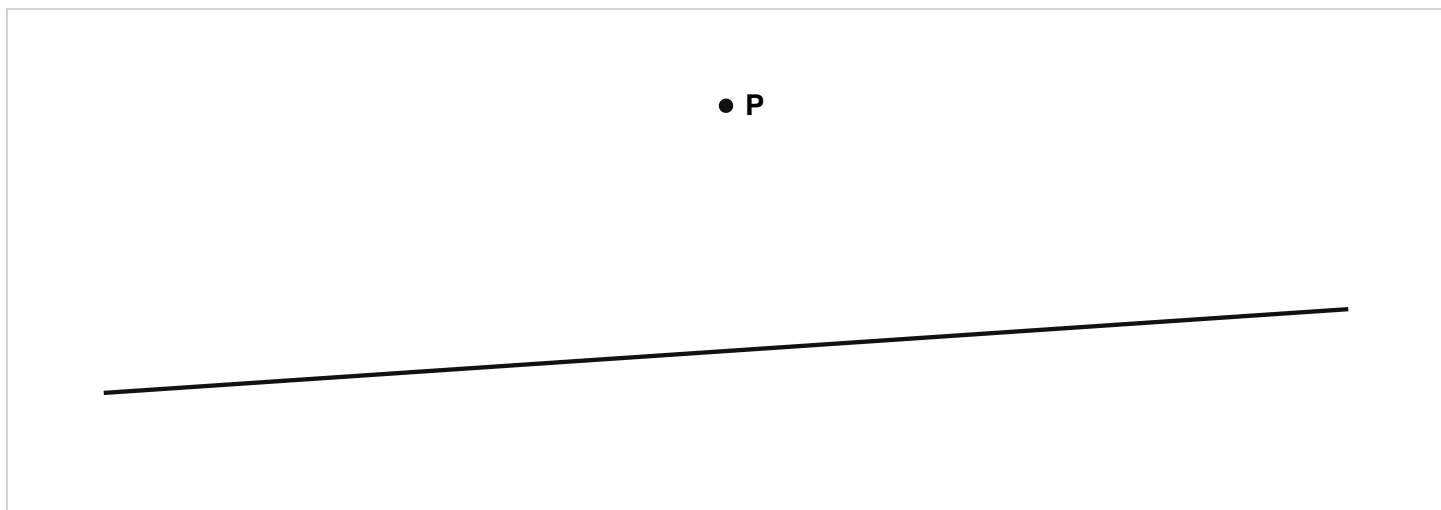
24. Construct a **copy of the angle** at V , with its vertex at W and one side along the new ray.



25. Construct the **angle bisector** of the angle at B.



26. Construct the line **perpendicular** to the given line through point P.



27. Starting at A, construct a **regular hexagon** inscribed in the circle with center O.

